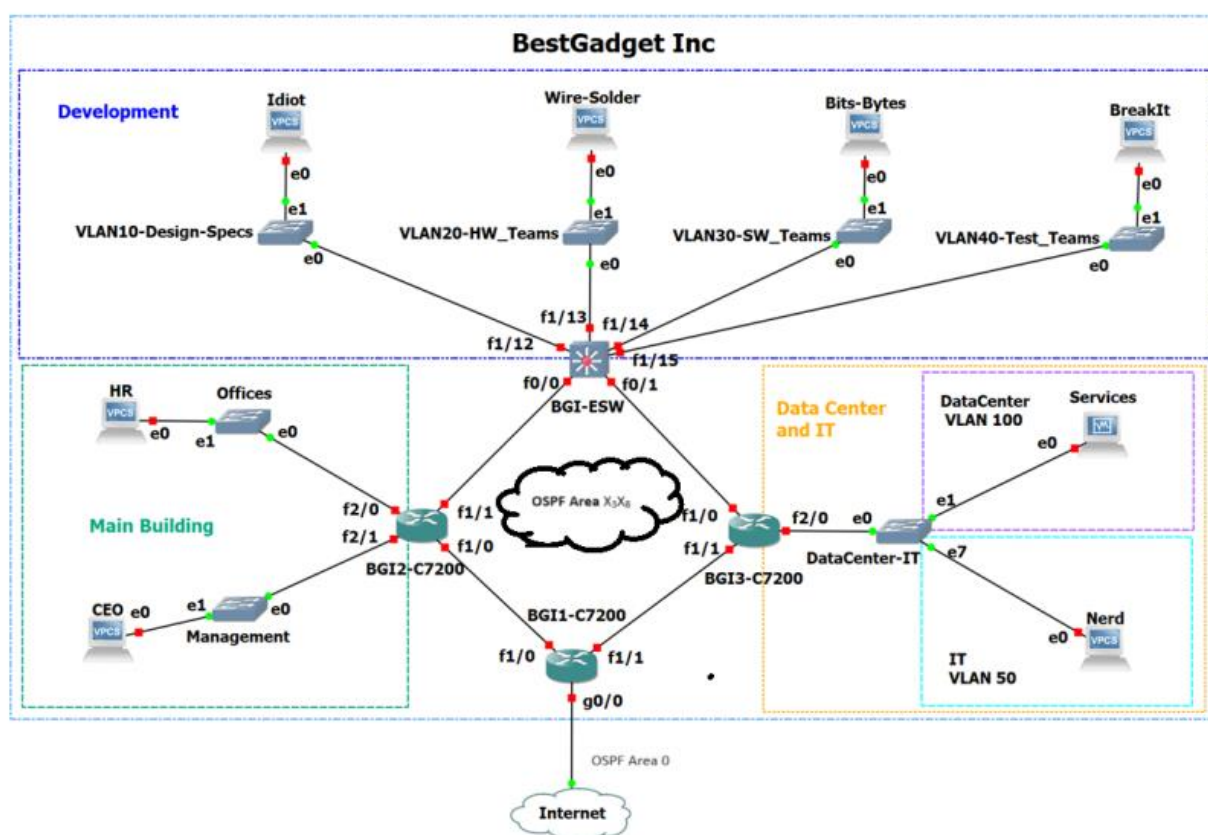


Redes de Telecomunicações

Project 2023/2024

Gonçalo Cristóvão [109041]
Eduardo Gonçalves [109435]

Addressing Charts



Começamos por substituir os valores:

Public IPv4 (sub)Network: 201.1X10X4.1X8X2.128/25

Private IPv4 Network: 172.2X6.1X42.0/23

Global IPv6 Network: 2002:AX1X5X7:BX9X8X2::/48

Student #1 University Number: 0/1 X1X2X3X4X5

Student #2 University Number: 0/1 X6X7X8X9X10

Para:

Public IPv4 (sub)Network: 201.154.149.128/25

Private IPv4 Network: 172.20.142.0/23

Global IPv6 Network: 2002:A019:B349::/48

Student #1 University Number: 0/1 09041

Student #2 University Number: 0/1 09435

Alterando cada "X" pelo número respetivo.

Public IPv4 Addressing (201.154.149.128/25)								
	Department	Network Address	Mask (no. of bits)	Broadcast Address (if applicable)	Available addresses for hosts and routers (first-last)	Used addresses for hosts (First-last) (if applicable)	Gateway #1 address (if applicable)	Gateway #2 address (if applicable)
NAT/PAT	X	201.154.149.128	27	201.154.149.159	201.154.149.129 - 201.154.149.158	201.154.149.129 - 201.154.149.149	X	
Data Center and IT	IT VLAN 50	201.154.149.224	28	201.154.149.239	201.154.149.225 - 201.154.149.238	201.154.149.225 - 201.154.149.235	201.154.149.238	
	DataCenter VLAN 100	201.154.149.160	27	201.154.149.191	201.154.149.161 - 201.154.149.190	201.154.149.161 - 201.154.149.177	201.154.149.190	
Main Building	Office	201.154.149.248	29	201.154.149.255	201.154.149.249 - 201.154.149.254	201.154.149.249 - 201.154.149.253	201.154.149.254	
	Management	201.154.149.208	28	201.154.149.223	201.154.149.209 - 201.154.149.222	201.154.149.209 - 201.154.149.221	201.154.149.222	
Development	VLAN10 – Design-Specs	201.154.149.240	29	201.154.149.247	201.154.149.241 - 201.154.149.246	201.154.149.241 - 201.154.149.245	201.154.149.246	
	VLAN20 – HW_Teams	X	X	X	X	X	X	
	VLAN30 – SW_Teams	X	X	X	X	X	X	
	VLAN40 – Test_Teams	201.154.149.192	28	201.154.149.207	201.154.149.193 - 201.154.149.206	201.154.149.193 - 201.154.149.206	201.154.149.206	

- NAT/PAT: 20 addresses: utilizamos uma máscara 27 pois o número máximo de endereços admitidos nesta máscara é 32.

- DataCenter VLAN: 16 addresses utilizamos uma máscara 28 pois o número máximo de endereços admitidos nesta máscara é 16.

- IT VLAN: 10 addresses utilizamos uma máscara 28 pois o número máximo de endereços admitidos nesta máscara é 16.

- Management: 12 addresses utilizamos uma máscara 28 pois o número máximo de endereços admitidos nesta máscara é 16.

- Offices: 4 addresses utilizamos uma máscara 28 pois o número máximo de endereços admitidos nesta máscara é 16

- Design-Specs VLAN: 5 addresses utilizamos uma máscara 28 pois o número máximo de endereços admitidos nesta máscara é 16.

- Test_Teams: 13 addresses utilizamos uma máscara 28 pois o número máximo de endereços admitidos nesta máscara é 16.

- Hardware_Teams: 0 addresses

- Software_Teams: 0 addresses

E as restantes máscaras, utilizamos o mesmo formato de endereçamento.

Private IPv4 Addressing (172.20.142.0/23)								
	Department	Network Address	Mask (no. of bits)	Broadcast Address (if applicable)	Available addresses for hosts and routers (first-last)	Used addresses for hosts (First-last) (if applicable)	Gateway #1 address (if applicable)	Gateway #2 address (if applicable)
Data Center and IT	IT VLAN 50	×	×	×	×	×	×	
	DataCenter VLAN 100	×	×	×	×	×	×	
Main Building	Office	172.20.143.128	26	172.20.143.191	172.20.143.129 - 172.20.143.190	172.20.143.129 - 172.20.143.190	172.20.143.190	
	Management	×	×	×	×	×	×	
Development	VLAN10 – Design-Specs	172.20.143.224	28	172.20.143.239	172.20.143.225 - 172.20.143.238	172.20.143.225 - 172.20.143.237	172.20.143.238	
	VLAN20 – HW_Teams	172.20.143.0	25	172.20.143.127	172.20.143.1 - 172.20.143.126	172.20.143.1 - 172.20.143.125	172.20.143.126	
	VLAN30 – SW_Teams	172.20.142.0	24	172.20.142.255	172.20.142.1 - 172.20.142.254	172.20.142.1 - 172.20.142.240	172.20.142.254	
	VLAN40 – Test_Teams	172.20.143.192	27	172.20.143.223	172.20.143.193 - 172.20.143.222	172.20.143.193 - 172.20.143.221	172.20.143.222	
Interconections	BGI – ESW <-> BGI3	172.20.143.240	30		172.20.143.241 - 172.20.143.243	172.20.143.241 - 172.20.143.242		
	BGI – ESW <-> BGI2	172.20.143.244	30		172.20.143.245 - 172.20.143.247	172.20.143.245 - 172.20.143.246		
	BGI1 <-> BGI3	172.20.143.248	30		172.20.143.249 - 172.20.143.251	172.20.143.249 - 172.20.143.250		
	BGI1 <-> BGI2	172.20.143.252	30		172.20.143.252 - 172.20.143.255	172.20.143.252 - 172.20.143.253		

Global IPv6 Addressing (2002:A019:B249:200::/48)							
	Department	Network Address	Mask (no. of bits)	Available addresses for hosts and routers (first-last)	Used addresses for hosts (First-last) (if applicable)	Gateway #1 address (if applicable)	Gateway #2 address (if applicable)
Data Center and IT	IT VLAN 50	2002:A019:B349:200::	64	*	*	2002:A019:B349:200:1::	
	DataCenter VLAN 100	2002:A019:B349:201::	64	*	*	2002:A019:B349:201:1::	
Main Building	Office	2002:A019:B349:100::	64	*	*	2002:A019:B349:100:1::	
	Management	2002:A019:B349:101::	64	*	*	2002:A019:B349:101:1::	
Development	VLAN10 – Design-Specs	2002:A019:B349::	64	*	*	2002:A019:B349:1::	
	VLAN20 – HW_Teams	2002:A019:B349:1::	64	*	*	2002:A019:B349:1:1::	
	VLAN30 – SW_Teams	2002:A019:B349:2::	64	*	*	2002:A019:B349:2:1::	
	VLAN40 – Test_Teams	2002:A019:B349:3::	64	*	*	2002:A019:B349:3:1::	
Interconnections	BGI – ESW <-> BGI3	2002:A019:B349:300::	126	2002:A019:B349:300::1 - 2002:A019:B349:300::3	2002:A019:B349:300::1 - 2002:A019:B349:300::2		
	BGI – ESW <-> BGI2	2002:A019:B349:300::4	126	2002:A019:B349:300::5 - 2002:A019:B349:300::7	2002:A019:B349:300::5 - 2002:A019:B349:300::6		
	BGI1 <-> BGI3	2002:A019:B349:300::8	126	2002:A019:B349:300::9 - 2002:A019:B349:300::B	2002:A019:B349:300::9 - 2002:A019:B349:300::A		
	BGI1 <-> BGI2	2002:A019:B349:300::C	126	2002:A019:B349:300::D - 2002:A019:B349:300::F	2002:A019:B349:300::D - 2002:A019:B349:300::E		

* All address within the range.

Fizemos este esquema para demonstrar as ligações efetuadas:

